ANSI/RVIA DC-2025

Standard for DC Voltage Systems in Recreational Vehicles

2025 Edition

Approved: October 17th, 2024

Sponsor: Recreational Vehicle Industry Association (RVIA)

ORIGIN AND DEVELOPMENT

Low voltage electrical standards for the conversion vehicle industry were non-existent prior to the development of this standard. The low voltage electrical requirements of the National Electrical Code (NEC) (ANSI/NFPA 70) found in Article 551 were used in the late 1980s and early 1990s. However, it was determined that NEC requirements did not allow the conversion vehicle manufacturer and the original chassis manufacturer to design and interface low voltage systems that were totally compatible.

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- **2025 Edition:** Retitled "Standard for DC Voltage Systems in Recreational Vehicles" to harmonize with emerging DC technologies. References to conversion vehicles were removed. New requirements for non-vented batteries were added in Section 2-3. The document was reorganized: previous Chapters 3, 4, 5, and 6 were condensed into Chapter 3 (Low Voltage Circuits); Chapter 4 contains new content on High Voltage Circuits; Chapter 6 covers Photovoltaic Systems.

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CHAPTER 1: GENERAL

1-1 Introduction

- **1-1.1 Need for Standard:** Developed to ensure uniformity and compatibility between RV manufacturers and OEM chassis electrical systems.
- **1-1.3 Scope:** This standard covers the installation of low voltage and high voltage direct current (DC) electrical systems and devices within recreational vehicles. It also covers additions, deletions, or modifications to the original equipment chassis manufacturer's electrical system.

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1-2 Not Covered This standard does not cover circuits used for signaling only, low-voltage circuits with a connected load of less than one ampere, portions covered by CFR 49 400-999, and circuits for items listed in Appendix 1.

1-3 Alternate Materials, Equipment, and Procedures Provisions are not intended to prevent the use of alternate materials or methods if acceptable to the authority having jurisdiction and substantiated by sufficient evidence of safety.

1-4 Limitations This standard is not intended as a design specification or an instruction manual.

1-5 Differing Standards Whenever recognized standards and this standard differ, the requirements of this standard shall apply.

1-6 U.S. Federal Regulations Federal regulations under the National Highway Traffic Safety Administration (NHTSA) may supersede all or part of this standard as applied to regulated motor vehicles.

1-7 Definitions

- **Authority Having Jurisdiction:** The organization, office, or individual responsible for approving equipment, an installation, or a procedure.
- **Battery, Auxiliary:** A device for storage of low voltage electrical energy installed by the RV manufacturer.
- **Battery, Source:** A device for storage of low voltage electrical energy installed by the original vehicle chassis manufacturer.
- **Battery Management System (BMS):** A battery control circuit with active protection that monitors cells to prevent overcharge, overcurrent, overtemperature, and overdischarge.
- **Converter:** A device that changes electrical energy from alternating current to direct current.
- **Ground:** The negative terminal of the battery or any conductor or metal current-carrying part of the vehicle chassis at the same potential.
- **High Voltage:** An electromotive force rated above 60 volts DC and up to 500 volts DC.
- **Low Voltage (LV):** An electromotive force rated at 60 volts DC or less.

- **Photovoltaic (PV) System:** The total components and subsystem that convert solar energy into electric energy.

CHAPTER 2: POWER SOURCES

2-1 Power Source Power source connections shall be accomplished at any of the following or by methods approved by the OEM:

1. Battery terminal or equal
2. Battery cable
3. Starter solenoid
4. OEM Factory-supplied power feed connectors
5. Converter
6. Charge line from tow vehicle
7. Photovoltaic (PV) system (Solar Panels)

2-2 OEM Circuit Extensions Additions to the OEM wiring system shall be permitted if all of the following conditions are met:

1. It is documented that the OEM's wiring system can safely support the additional amperage.
2. The OEM installed overcurrent protection is not replaced with one of a higher amperage.
3. Conductors meet or exceed the gauge and temperature ratings of the OEM conductor.

2-3 Auxiliary Battery Installations Compartments, hoods, or housings identified for housing batteries (e.g., by the presence of battery cables) shall meet the following requirements.

2-3.1 Vented Battery Installations (e.g., Lead Acid)

- **2-3.1.1 Location:** Shall not be installed in compartments containing spark or flame-producing equipment, except in engine/generator compartments if the only charging source is from that engine.

- **2-3.1.2 Vapor Resistance:** Compartments shall be vapor resistant to the interior and ventilated directly to the exterior.
- **2-3.1.3 Ventilation:** Compartments shall have openings with a minimum area of 1.7 in^2 (1097 mm^2) within 2 in. (50.8 mm) of both the top and bottom.
- **2-3.1.6 Securement:** Batteries shall be secured to withstand a load equal to eight times the battery system's weight applied to its center of gravity.

2-3.2 Non-vented Battery Installations (e.g., Lithium-ion)

- **2-3.2.1 BMS:** Battery systems shall include a BMS.
- **2-3.2.2 Listing:** Systems ≤ 60V DC shall be listed to UL 2054, IEC/UL 62133-2, or UL 1973. Systems > 60V DC shall be listed to UL 1973 or recognized motive standards (e.g., UL 2580).

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- **2-3.2.3 Vented Battery Prevention:** Compartments shall be designed such that they are incapable of installing a vented battery, or the system shall be incapable of electrically operating a vented battery.
- **2-3.2.7 Safety Label:** A warning label must be affixed to the battery system or compartment stating:

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WARNING: This battery area/compartment is designed for the use of non-vented batteries only. Do not place a battery that requires venting into this area/compartment. Vented batteries can release poisonous and flammable gases. Can lead to a fire or explosion and result in death or serious injury.

2-4 Auxiliary Battery Grounding The chassis-grounding terminal of the battery shall be grounded to the vehicle chassis with a minimum No. 8 AWG copper conductor or equivalent. If the power lead exceeds No. 8 AWG, the grounding conductor shall be of equal size.

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2-5 Converters

- **Listing:** All converters and transformers shall be listed for use in recreational vehicles.

- **AC Transformers:** Circuits fed from AC transformers shall not supply DC appliances.
- **2-5.1 Bonding:** The metal enclosure of the converter shall be bonded to the vehicle frame with a minimum No. 8 copper conductor.

2-6 Low Voltage Solar Prep Installations

Permitted if:

1. Conductors have overcurrent protection.
2. Controller location is identified.
3. Conductors have no exposed ends and are retained in the location.
4. Conductors are labeled "Battery +"/"Battery -" and "PV +"/"PV -" (or "Solar").

2-7 Low Voltage Solar Installations Solar controllers shall be listed, negatively grounded, and rated for no less than the maximum array voltage and current (array current x 1.25).

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CHAPTER 3: LOW VOLTAGE CIRCUITS

3-1 Circuit Protection

- **3-1.1 General:** All conductors shall be provided with overcurrent protection.
    - *Exception 1:* Solar panel circuits do not require protection if conductors have sufficient ampacity.
    - *Exception 2:* Braking, cranking, regulated exterior lights, and pigtails <10 inches are exempt.
- **3-1.2 Sizing:** Protection shall not exceed the ampacity of conductors shown in Tables 1, 2, or 3.
- **3-1.5 Location:** Protection device shall be accessible and within 18 in. (457 mm) of the power source connection.

Table 1: Maximum Allowable Ampacity of Conductors

(No Wire Bundling Restrictions, Temperature Rating 90°C or Greater)

Wire Size (AWG)	Ampacity
20	3
18	6
16	8
14	15
12	20
10	30
8	40
6	55
4	75
2	100
1	110
1/0	125
2/0	150
3/0	175
4/0	200

Table 2: Maximum Allowable Ampacity (Single Conductor, Not Bundled)

(Partial table for 90°C, 105°C, 125°C)

AWG	90°C	105°C	125°C
20	5	7.5	7.5
18	20	20	25
16	25	30	35
14	35	40	45
12	45	50	60
10	60	70	80

Table 3: Correction Factors for Bundling

(Applied to Table 2)

Number of Conductors	Factor
2 to 3	0.70
4 to 6	0.60
7 to 24	0.50
25 and above	0.40

3-1.3 Types: Circuit breakers or fuses shall be of an approved type. Circuit breakers shall be trip-free. **3-1.4 Marking:** Fuseholders marked with max replacement size; breakers marked with amperage rating.

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3-2 Low Voltage Conductors

- **3-2.1 Materials:** Copper conductors, stranded type.
- **3-2.2 Types:** Shall conform to SAE J1127/J1128 or NEC Table 310-4. Minimum insulation rating 90°C (195°F). Conductors in engine compartments or <10 inches from exhaust must be rated 125°C (257°F).

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- **3-2.3 Identification:** Listed conductors marked by agency; SAE marked by SAE requirements; others marked with temp/gauge/material .
- **3-2.4 Circuit Boards:** Printed circuit boards used in place of wire shall be identified and marked regarding maximum ampacity. Printed circuit boards shall be used within their stated ratings.

Here is the full, cleaned text from **Scanned Document 3.pdf**. This picks up where Document 2 left off, completing Chapter 3 and covering the remainder of the standard.

CHAPTER 3: LOW VOLTAGE CIRCUITS (Continued)

3-2.5 Grounded Circuit Identification All ground returns to chassis or power source shall be uniquely identified as separate from the supply conductors (i.e., color, marking, etc.).

3-2.6 Other Circuit Identification Each circuit conductor shall maintain its established identification throughout its run.

3-3 Low Voltage Conductor Protection, Routing, and Securing

3-3.1 Conductor Protection Conductors shall be protected against physical damage and shall be supported. Where insulated conductors are attached to the structure, the conductor insulation shall be supplemented by an additional wrap or layer of equivalent material, except that jacketed cables shall not be required to be so protected. Wiring shall be routed away from sharp edges, moving parts, or heat sources.

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3-3.2 Conductor Routing Direct-current circuits shall be physically separated by at least a 1/2 in. (12.7 mm) gap or other approved means from 120/240 volt circuits. Acceptable methods shall be by clamping, routing, or equivalent means that ensure permanent total

separation. The external jacket of the 120/240-volt circuit nonmetallic-sheathed cables and 120/240 volt circuit flexible cords shall be permitted as adequate separation.

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- **3-3.2.1 Harness:** Conductors shall be grouped in harness form where required by the OEM.
- **3-3.2.2 Conductor Continuity:** Conductor insulation shall be continuous between terminals.

3-3.3 Conductor Securing

- **3-3.3.1 Strain Relief:** Conductors connecting components that can move with relation to each other shall be installed with a loop, slack, or other means of strain relief.
- **3-3.3.2 Accessibility:** At least 4 in. (101.6 mm) of free conductor shall be provided after the last point of conductor securement to permit service and accessibility of each device.
- **3-3.3.3 Conductor Support:** Conductor(s) shall be supported and secured at intervals not exceeding 4 1/2 ft. (1.4 m).

3-4 Low Voltage Connections

3-4.1 Conductor Splicing and Terminals

- **3-4.1.1 Splicing:** Conductors shall be spliced or properly joined with connectors or by soldering.
- **3-4.1.2 Tensile Force:** Each conductor splice shall be able to withstand a tensile force equal to at least the value shown in Table 4 for a one-minute duration and not break or separate.

Table 4: Tensile Strength Values for Connectors

(Pounds of Force)

Wire Size (AWG)	**UL-486A (Terminals)**	**UL-486C (Splices/Wire Nuts)**	**UL-310 (Quick Disconnects)**
20	13	10	13

Wire Size (AWG)	UL-486A (Terminals)	UL-486C (Splices/Wire Nuts)	UL-310 (Quick Disconnects)
18	20	10	20
16	30	15	30
14	50	25	50
12	70	35	70
10	80	40	80
8	90	45	N/A
6	100	50	N/A

- **3-4.1.3 Size and Use:** All connectors and terminals shall be identified for the conductor size and type.
- **3-4.1.4 Soldered Splices:** Shall be twisted or mechanically secured before soldering.
- **3-4.1.6 Wire Nuts:** Twist-on wire nuts, if used, shall be listed and used within the terms of their listing.
- **3-4.1.8 Protection:** All splices and joints shall be covered with an insulation equivalent to that of the conductors. Free ends shall be protected from accidental shorting.

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- **3-4.1.14 Stacking:** No more than four terminals shall be secured to any one terminal stud.

3-4.2 Grounding

- **3-4.2.1 Attachment Material:** Screws, nuts, bolts, and lock washers shall be cadmium, tin, zinc, or equivalently plated.

- **3-4.2.5 Electrical Connections:** The surface shall be cleaned and free from oxide or paint, or the terminal shall be connected through use of a serrated paint-cutting terminal or lockwasher.

CHAPTER 4: HIGH VOLTAGE CIRCUITS

4-1 Circuit Protection

- **4-1.1 General:** All conductors shall be provided with overcurrent protection.
- **4-1.2 Sizing:** Protection shall not exceed the ampacity of conductors shown in Tables 1, 2, or 3.

4-2 Conductors

- **4-2.1 Materials:** Copper conductors, stranded type.
- **4-2.3 Color:**
    - **4-2.3.1:** Individual conductors not in a harness shall be **orange**. (e.g., positive orange, negative orange w/ black stripe) .
    - **4-2.3.2:** Harnesses containing high voltage shall be visually identified with a permanent **orange** harness covering material.

4-3 Conductor Protection, Routing and Securing

- **4-3.1.3 Sharp Edges:** Conductors passing within 1 in. (25 mm) of holes or edges must be protected (tubing/conduit) or the edge must be free of sharp edges .
- **4-3.2 Routing:** Must be physically separated by at least 1/2 in. (12.7 mm) from 120/240 volt circuits.
- **4-3.3 Securing:**
    - **4-3.3.2:** Secured every 24 in. (600mm) or less.
    - **4-3.3.3:** Bend radius less than five times the diameter of the conductor is not permitted.

4-4 Connectors Connectors shall be: (1) Listed for high voltage use; (2) Polarized; (3) Non-interchangeable with other systems; (4) Identical brand/compatible; (5) Latching or locking; (6) Guarded; (7) Sized for the circuit .

4-5 Marking

A symbol (yellow with black lightning bolt arrow) shall be present on/near electric energy storage devices and protection barriers exposing live parts .

CHAPTER 5: EQUIPMENT

5-1 General Installation shall meet standard requirements and manufacturer's instructions. If they differ, this standard applies (unless the device is listed, then listing terms apply) .

5-2 Devices

- **5-2.1 Switches and Relays:** Must be rated for system voltage and connected load. Ratings for tungsten lights and inductive loads (motors) shall be **150%** of the connected load or as specified by the manufacturer .
- **5-2.2 Cigarette Lighters:** Receptacles must be installed in a non-combustible outlet box or be thermally protected.

5-3 Utilization Equipment

- **5-3.1 Lighting Fixtures:** Interior low-voltage fixtures shall be listed (Exception: fixtures <4 watts) .
    - **5-3.1.1 Movable Beds:** Fixtures must be de-energized if the lens/bulb comes within 1 in. (25.4 mm) of a movable bed/section with a flame-spread index of 26 or more.
    - **5-3.1.2 Mattresses:** Fixtures shall not be installed within 18 in. directly above a mattress unless permitted by the listing or shielded.

CHAPTER 6: HIGH VOLTAGE PHOTOVOLTAIC SYSTEMS

6-1 General Requirements

- **6-1.1 Equipment:** DC combiners, PV source circuits, charge controllers, and disconnects must be listed.
- **6-1.3 Qualified Persons:** Installation involving high voltage shall be performed only by qualified persons.
- **6-1.4 Locations:** PV equipment shall not be installed in bathrooms.

6-2 Circuit Requirements

- **6-2.1 Maximum Voltage:** Shall be no greater than **500 volts** DC.
- **6-2.1.2 Maximum Current:** Shall be no greater than **30 amperes**.

6-3 Overcurrent Protection Required when conductor ampacity is not sufficient to carry max current x 125%. Protection must be located within 18 in. (457.2 mm) of the source.

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6-3.3 PV Array Emergency Disconnect A disconnect must be provided to disconnect the array from the RV during an emergency.

- **6-3.3.1 Location:** On the roof within 18 in. (457.2 mm) from the point of entry.
- **6-3.3.4 Marking:** Permanent warning label on the exterior near the disconnect.

WARNING: EMERGENCY HIGH VOLTAGE DC PV SYSTEM DISCONNECT

6-3.4 PV Array Service Disconnect A disconnect must be provided to disconnect the array from the charge controller during maintenance.

- **6-3.4.2 Location:** Same location as the battery disconnect.
- **6-3.4.4 Marking:** Permanent notice label.

NOTICE: PV SYSTEM SERVICE DISCONNECT

6-4 Conductors High voltage conductors shall be orange or marked with orange. Outdoor conductors can be PV wire or listed for sunlight resistance .

6-5.4 Charge Controllers Must be listed and negatively grounded. Must have a label adjacent to guarded high voltage connections:

WARNING: HIGH VOLTAGE DC - DISCONNECT SOLAR ARRAY BEFORE REMOVING GUARD .

CHAPTER 7: TESTING

7-1 Operational Test An operational test of all direct current circuits shall be conducted to demonstrate that all equipment is connected and in electrical working order. This test shall be performed after all production activities that may damage conductors (fasteners, hole cutting) are completed .

CHAPTER 8: REFERENCED PUBLICATIONS

Includes standards from:

- **SAE:** J553, J1127 (Battery Cable), J1128 (Low Voltage Primary Cable), etc. .
- **UL:** UL 486A (Wire Connectors), UL 1973 (Batteries), UL 2054, etc. .
- **NFPA:** NFPA 70 (National Electrical Code).
- **ASTM:** ASTM E84 (Surface Burning Characteristics).

APPENDICES

Appendix 1: Systems Not Covered

1. Speed control
2. Windshield wiping
3. Windshield defrosting
4. Headlamp systems
5. Hood latching
6. Braking
7. Steering
8. Throttle control
9. Transmission neutral start

Appendix 2: Standard Ampere Ratings (Fuses/Breakers) Standard ratings include: 1, 3, 5, 7.5, 10, 15, 20, 25, 30, 40, 50, 60, 80, 100, 125, 150, 175, 200, 225, 300, 400, 500.

Appendix 3: Conductor Sizing Table (12VDC @ 10% Voltage Drop)

(Partial data extracted from table)

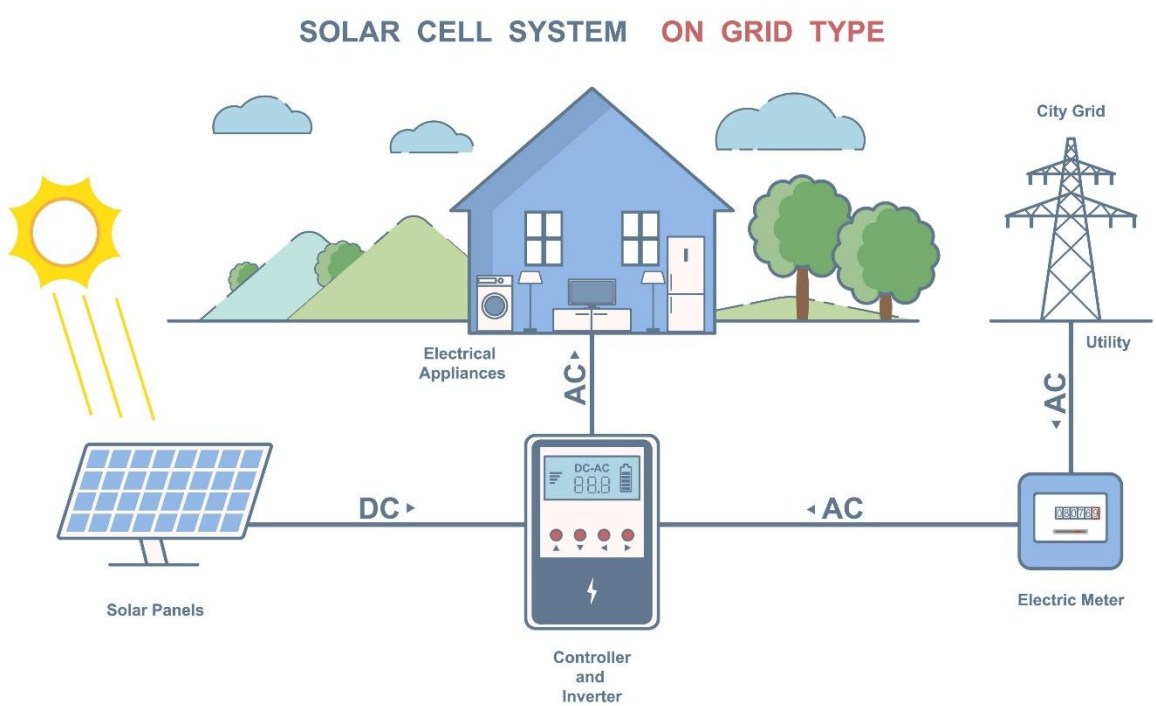
Current (Amps)	**14 AWG Max Length (ft)**	**10 AWG Max Length (ft)**	**6 AWG Max Length (ft)**
5	23	58	147

Current (Amps)	14 AWG Max Length (ft)	10 AWG Max Length (ft)	6 AWG Max Length (ft)
10	11	29	73
15	8	19	49
20	6	14	37
30	4	9	24

Appendix 4: Recreational Vehicle (RV) Definitions

- **Motorhome:** Type A (Check chassis), Type B (Van), Type C (Cut-away).
- **Towable:** Fifth-Wheel, Folding Camping Trailer, Travel Trailer, Truck Camper.

Appendix 5: Photovoltaic (PV) System Components



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Explore

(Diagram illustrates flow: PV Cells -> Module -> String -> Array -> DC Combiner -> PV System Disconnect -> Charge Controller -> Energy Storage System).